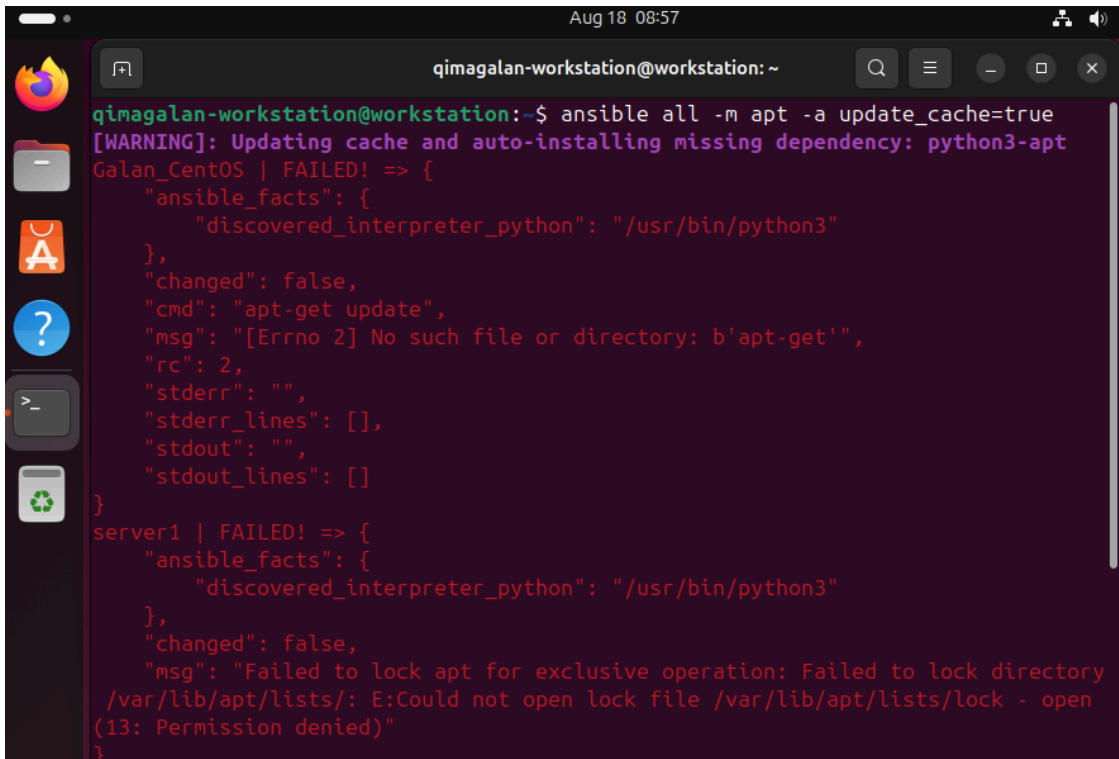


COURSE CODE/TITLE:	CPE 212 - Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	08/18/2026	DATE SUBMITTED:	08/24/2026
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ACTIVITY NO. 4 Ansible Basic			
1. PROCEDURE OUTPUT			

Task 1: Run elevated ad hoc commands

- Locally, we use the command `sudo apt update` when we want to download package information from all configured resources. The sources are often defined in `/etc/apt/sources.list` file and other files located in `/etc/apt/sources.list.d/` directory. So, when you run the update command, it downloads the package information from the Internet. It is useful to get info on an updated version of packages or their dependencies. We can only run an apt update command in a remote machine. Issue the following command:

`ansible all -m apt -a update_cache=true`



```

qimagalan-workstation@workstation:~$ ansible all -m apt -a update_cache=true
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
Galan_CentOS | FAILED! => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "cmd": "apt-get update",
  "msg": "[Errno 2] No such file or directory: b'apt-get'",
  "rc": 2,
  "stderr": "",
  "stderr_lines": [],
  "stdout": "",
  "stdout_lines": []
}
server1 | FAILED! => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "msg": "Failed to lock apt for exclusive operation: Failed to lock directory
/var/lib/apt/lists/: E:Could not open lock file /var/lib/apt/lists/lock - open
(13: Permission denied)"
}

```

What is the result of the command? Is it successful?

No. It throws an error for the Ubuntu servers, stating that it fails to create a lock for apt since it does not have any permission.

Try editing the command and add something that would elevate the privilege. Issue the command `ansible all -m apt -a update_cache=true --become --ask-become-pass`. Enter the sudo password when prompted. You will notice now that the output of this command is a success. The `update_cache=true` is the same thing as running `sudo apt update`. The `--become` command elevates the privileges and the `--ask-become-pass` asks for the password. For now, even if we only changed the packaged index, we were able to change something on the remote server. You may notice after the second command was executed, the status is

CHANGED compared to the first command, which is **FAILED**.

```
qimagalan-workstation@workstation:/etc/ansible
qimagalan-workstation@workstation:/etc/ansible$ ansible all -m apt -a update_cache=true -b -K -i inventory.yaml
BECOME password:
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
Galan_CentOS | FAILED! => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "cmd": "apt-get update",
  "msg": "[Errno 2] No such file or directory: b'apt-get'",
  "rc": 2,
  "stderr": "",
  "stderr_lines": [],
  "stdout": "",
  "stdout_lines": []
}
```

```
server1 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "cache_update_time": 1787016309,
  "cache_updated": true,
  "changed": true
}
server2 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "cache_update_time": 1787016310,
  "cache_updated": true,
  "changed": true
}
qimagalan-workstation@workstation:/etc/ansible$
```

2. Let's try to install VIM, which is an almost compatible version of the UNIX editor Vi. To do this, we will just change the module part in 1.1 instruction. Here is the command: ***ansible all -m apt -a name=vim-nox --become --ask-become-pass***. The command would take some time after typing the password because the local machine instructed the remote servers to actually install the package.

```
qimagalan-workstation@workstation: /etc/ansible
qimagalan-workstation@workstation: /etc/an... x qimagalan-workstation@workstation: /etc/an... x
qimagalan-workstation@workstation: /etc/ansible$ ansible all -m apt -a name=vim-nox
-b -K -i inventory.yaml
BECOME password:
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
qimagalan-CentOS | FAILED! => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "cmd": "apt-get update",
  "msg": "[Errno 2] No such file or directory: b'apt-get'",
  "rc": 2,
  "stderr": "",
  "stderr_lines": [],
  "stdout": "",
  "stdout_lines": []
}
server1 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "cache_update_time": 1787016309,
  "cache_updated": false,
  "changed": true,
```

```
qimagalan-workstation@workstation: /etc/ansible
qimagalan-workstation@workstation: /etc/an... x qimagalan-workstation@workstation: /etc/an... x
]
server2 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "cache_update_time": 1787016310,
  "cache_updated": false,
  "changed": true,
  "stderr": "",
  "stderr_lines": [],
  "stdout": "Reading package lists...\nBuilding dependency tree...\nReading st
ate information...\nThe following additional packages will be installed:\n font
s-lato javascript-common libjs-jquery liblua5.1-0 libruby libruby3.2\n rake rub
y ruby-net-telnet ruby-rubygems ruby-sdbm ruby-webrick ruby-xmlrpc\n ruby3.2 ru
bygems-integration vim-common vim-runtime vim-tiny xxd\nSuggested packages:\n a
pache2 | lighttpd | httpd ri ruby-dev bundler cscope vim-doc indent\nThe followi
ng NEW packages will be installed:\n fonts-lato javascript-common libjs-jquery
liblua5.1-0 libruby libruby3.2\n rake ruby ruby-net-telnet ruby-rubygems ruby-s
dbm ruby-webrick ruby-xmlrpc\n ruby3.2 rubygems-integration vim-nox vim-runtime
\nThe following packages will be upgraded:\n vim-common vim-tiny xxd\n3 upgrade
d, 17 newly installed, 0 to remove and 141 not upgraded.\nNeed to get 19.7 MB of
archives.\nAfter this operation, 83.8 MB of additional disk space will be used.
\nGet:1 http://ph.archive.ubuntu.com/ubuntu noble/main amd64 fonts-lato all 2.01
```

a. Verify that you have installed the package in the remote servers. Issue the command *which vim* and the command *apt search vim-nox* respectively. Was the command successful?

```
qimagalan-workstation@server1: ~
qimagalan-workstation@workstation: ~ x qimagalan-workstation@server1: ~ x
qimagalan-workstation@server1:~$ which vim
/usr/bin/vim
qimagalan-workstation@server1:~$
```

```
qimagalan-workstation@server2: ~  
qimagalan-workstation@workstation: ~ x qimagalan-workstation@server2: ~  
qimagalan-workstation@server2:~$ which vim  
/usr/bin/vim  
qimagalan-workstation@server2:~$
```

```
Aug 18 09:37  
qimagalan-workstation@server1: ~  
qimagalan-workstation@server1: ~ x qimagalan-workstation@workstation: /etc/an...  
qimagalan-workstation@server1:~$ apt search vim-nox  
Sorting... Done  
Full Text Search... Done  
vim-nox/noble-updates,noble-security,now 2:9.1.0016-1ubuntu7.18 amd64 [installed  
]  
  Vi IMproved - enhanced vi editor - with scripting languages support  
vim-tiny/noble-updates,noble-security,now 2:9.1.0016-1ubuntu7.18 amd64 [installe  
d,automatic]  
  Vi IMproved - enhanced vi editor - compact version  
qimagalan-workstation@server1:~$
```

```
Aug 18 09:38
qimagalan-workstation@server2: ~
qimagalan-workstation@server2: ~ x qimagalan-workstation@workstation: /etc/an... x v
qimagalan-workstation@server2:~$ apt search vim-nox
Sorting... Done
Full Text Search... Done
vim-nox/noble-updates,noble-security,now 2:9.1.0016-1ubuntu7.18 amd64 [installed]
  Vi IMproved - enhanced vi editor - with scripting languages support
vim-tiny/noble-updates,noble-security,now 2:9.1.0016-1ubuntu7.18 amd64 [install
d,automatic]
  Vi IMproved - enhanced vi editor - compact version
qimagalan-workstation@server2:~$
```

We will see that which vim shows a binary executable for vim which means it is installed. The apt search command also shows an '[installed]' status for both vim-nox and vim-tiny.

- b. Check the logs in the servers using the following commands: **cd /var/log**. After this, issue the command **ls**, go to the folder **apt** and open **history.log**. Describe what you see in the history.log.

```
Aug 18 09:41
qimagalan-workstation@server1: /var/log/apt
qimagalan-workstation@server1: /var/log/apt x qimagalan-workstation@workstation: /etc/an... x v
qimagalan-workstation@server1:~$ cd /var/log/apt
qimagalan-workstation@server1:/var/log/apt$ ls
eipp.log.xz history.log term.log
qimagalan-workstation@server1:/var/log/apt$ less history.log
qimagalan-workstation@server1:/var/log/apt$
```

```

Aug 18 09:40
qimagalan-workstation@server1: /var/log/apt
qimagalan-workstation@server1: /var/log/apt x qimagalan-workstation@workstation: /etc/an... x
Commandline: /usr/bin/apt-get -y -o Dpkg::Options::=--force-confdef -o Dpkg::Options::=--force-confold install vim-nox=2:9.1.0016-1ubuntu7.18
Requested-By: qimagalan-workstation (1000)
Install: ruby-sdbm:amd64 (1.0.0-5build4, automatic), liblua5.1-0:amd64 (5.1.5-9b
uild2, automatic), fonts-lato:amd64 (2.015-1, automatic), libruby:amd64 (1:3.2~u
buntu1, automatic), ruby-net-telnet:amd64 (0.2.0-1, automatic), rubygems-integra
tion:amd64 (1.18, automatic), libruby3.2:amd64 (3.2.3-1ubuntu0.24.04.8, automati
c), rake:amd64 (13.0.6-3, automatic), vim-nox:amd64 (2:9.1.0016-1ubuntu7.18), ru
by:amd64 (1:3.2~ubuntu1, automatic), vim-runtime:amd64 (2:9.1.0016-1ubuntu7.18,
automatic), ruby3.2:amd64 (3.2.3-1ubuntu0.24.04.8, automatic), libjs-jquery:amd6
4 (3.6.1+dfsg+~3.5.14-1, automatic), ruby-rubygems:amd64 (3.4.20-1, automatic),
javascript-common:amd64 (11+nmu1, automatic), ruby-xmlrpc:amd64 (0.3.2-2, automa
tic), ruby-webrick:amd64 (1.8.1-1ubuntu0.2, automatic)
Upgrade: xxd:amd64 (2:9.1.0016-1ubuntu7.17, 2:9.1.0016-1ubuntu7.18), vim-common:
amd64 (2:9.1.0016-1ubuntu7.17, 2:9.1.0016-1ubuntu7.18), vim-tiny:amd64 (2:9.1.00
16-1ubuntu7.17, 2:9.1.0016-1ubuntu7.18)
End-Date: 2026-08-18 09:29:15

Start-Date: 2026-08-18 09:31:55
Commandline: apt install python3-pip -y
Requested-By: qimagalan-workstation (1000)
Install: libgcc-13-dev:amd64 (13.3.0-6ubuntu2~24.04.1, automatic), libstdc++-13-
dev:amd64 (13.3.0-6ubuntu2~24.04.1, automatic), libalgorithm-merge-perl:amd64 (0
:

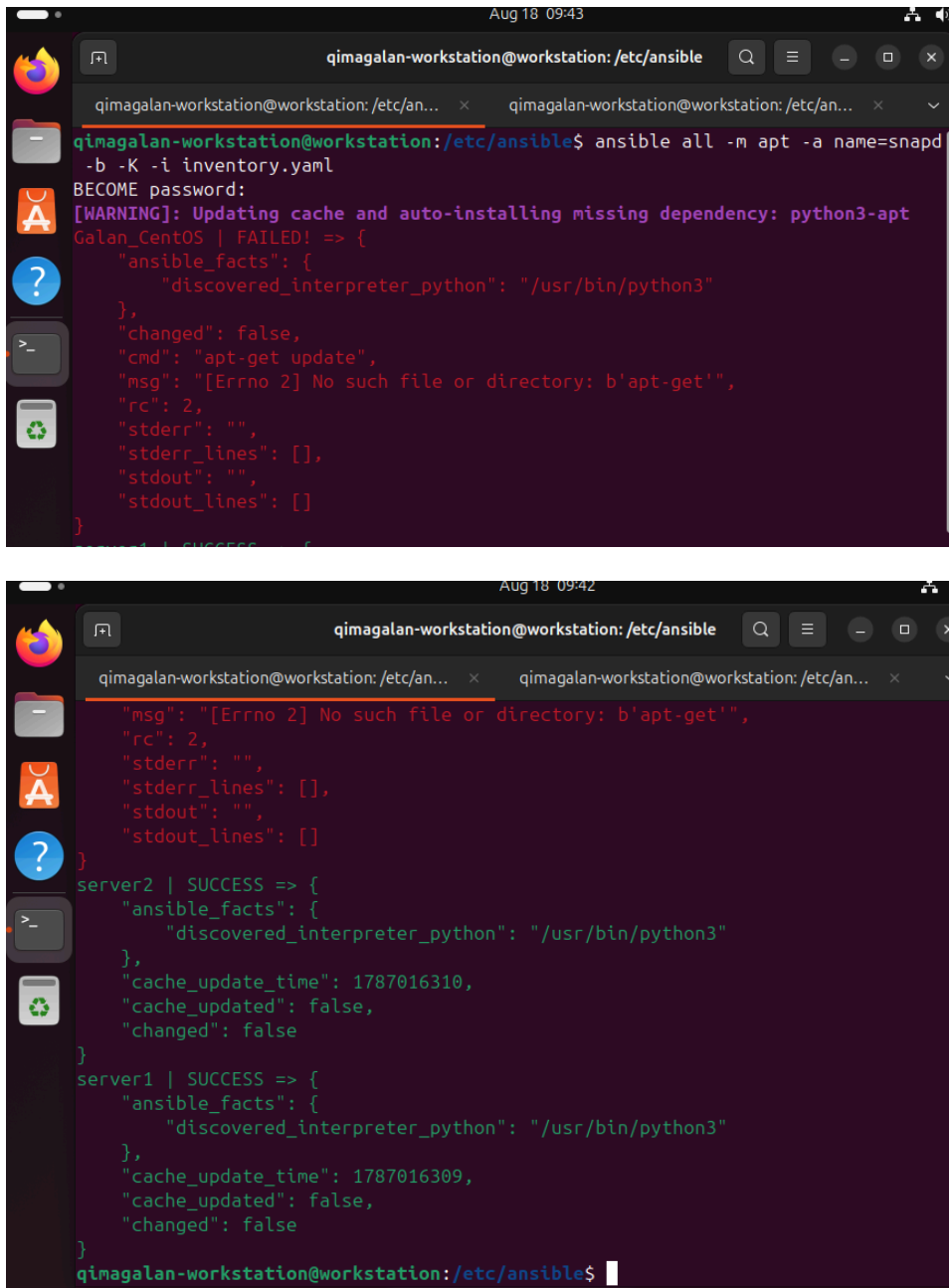
```

3. This time, we will install a package called snapd. Snap is pre-installed in the Ubuntu system. However, our goal is to create a command that checks for the latest installation package.

- a. Issue the command: `ansible all -m apt -a name=snapd --become --ask-become-pass`

Can you describe the result of this command? Is it a success? Did it change anything in the remote servers?

Yes, it succeeded for both server1 and server2 since they are both Ubuntu. However, based on the listed statuses, changed=false. Therefore, nothing has actually changed in the servers.



The image consists of two screenshots of a terminal window. The top screenshot shows the execution of the command `ansible all -m apt -a name=snapd -b -K -i inventory.yaml`. It prompts for a password and then displays a warning about updating the cache and installing the missing dependency `python3-apt`. The output for the `Galan_CentOS` host shows a failure with the message `[Errno 2] No such file or directory: b'apt-get'`. The bottom screenshot shows the output for the `server2` and `server1` hosts, both of which were successful in updating the package cache.

```
Aug 18 09:43
qimagalan-workstation@workstation: /etc/ansible
qimagalan-workstation@workstation: /etc/an... x qimagalan-workstation@workstation: /etc/an... x
qimagalan-workstation@workstation: /etc/ansible$ ansible all -m apt -a name=snapd
-b -K -i inventory.yaml
BECOME password:
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
Galan_CentOS | FAILED! => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "cmd": "apt-get update",
  "msg": "[Errno 2] No such file or directory: b'apt-get'",
  "rc": 2,
  "stderr": "",
  "stderr_lines": [],
  "stdout": "",
  "stdout_lines": []
}
server2 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "cache_update_time": 1787016310,
  "cache_updated": false,
  "changed": false
}
server1 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "cache_update_time": 1787016309,
  "cache_updated": false,
  "changed": false
}
qimagalan-workstation@workstation: /etc/ansible$
```

c. Now, try to issue this command: `ansible all -m apt -a "name=snapd state=latest" --become --ask-become-pass`

Aug 18 09:48

qimagalan-workstation@workstation: /etc/ansible

qimagalan-workstation@workstation: /etc/an... x qimagalan-workstation@workstation: /etc/an... x

```
qimagalan-workstation@workstation:/etc/ansible$ ansible all -m apt -a "name=snd
d state=latest" -b -K -i inventory.yaml
BECOME password:
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
Galan_CentOS | FAILED! => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "cmd": "apt-get update",
  "msg": "[Errno 2] No such file or directory: b'apt-get'",
  "rc": 2,
  "stderr": "",
  "stderr_lines": [],
  "stdout": "",
  "stdout_lines": []
}
server2 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "cache_update_time": 1787016310,
  "cache_updated": false,
  "changed": true,
```

```
Aug 18 09:48
qimagalan-workstation@workstation: /etc/ansible

qimagalan-workstation@workstation: /etc/an... x qimagalan-workstation@workstation: /etc/an... x v

"stdout_lines": []
}
server2 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "cache_update_time": 1787016310,
  "cache_updated": false,
  "changed": true,
  "stderr": "",
  "stderr_lines": [],
  "stdout": "Reading package lists...\nBuilding dependency tree...\nReading st
ate information...\nThe following packages will be upgraded:\n  snapd\n1 upgrade
d, 0 newly installed, 0 to remove and 140 not upgraded.\nNeed to get 36.4 MB of
archives.\nAfter this operation, 2761 kB of additional disk space will be used.\
nGet:1 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 snapd amd64
2.76+ubuntu24.04.1 [36.4 MB]\nFetched 36.4 MB in 21s (1726 kB/s)\n(Reading datab
ase ... \r(Reading database ... 5%\r(Reading database ... 10%\r(Reading database
... 15%\r(Reading database ... 20%\r(Reading database ... 25%\r(Reading databas
e ... 30%\r(Reading database ... 35%\r(Reading database ... 40%\r(Reading databa
se ... 45%\r(Reading database ... 50%\r(Reading database ... 55%\r(Reading datab
ase ... 60%\r(Reading database ... 65%\r(Reading database ... 70%\r(Reading datab
ase ... 75%\r(Reading database ... 80%\r(Reading database ... 85%\r(Reading datab
ase ... 90%\r(Reading database ... 95%\r(Reading database ... 100%\r(Reading d
```

```
Aug 18 09:48
qimagalan-workstation@workstation: /etc/ansible

qimagalan-workstation@workstation: /etc/an... x qimagalan-workstation@workstation: /etc/an... x v

"Processing triggers for dbus (1.14.10-4ubuntu4.1) ...",
"Processing triggers for desktop-file-utils (0.27-2build1) ..."
]
}
server1 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "cache_update_time": 1787016309,
  "cache_updated": false,
  "changed": true,
  "stderr": "",
  "stderr_lines": [],
  "stdout": "Reading package lists...\nBuilding dependency tree...\nReading st
ate information...\nThe following packages will be upgraded:\n  snapd\n1 upgrade
d, 0 newly installed, 0 to remove and 140 not upgraded.\nNeed to get 36.4 MB of
archives.\nAfter this operation, 2761 kB of additional disk space will be used.\
nGet:1 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 snapd amd64
2.76+ubuntu24.04.1 [36.4 MB]\nFetched 36.4 MB in 18s (2041 kB/s)\n(Reading datab
ase ... \r(Reading database ... 5%\r(Reading database ... 10%\r(Reading database
... 15%\r(Reading database ... 20%\r(Reading database ... 25%\r(Reading databas
e ... 30%\r(Reading database ... 35%\r(Reading database ... 40%\r(Reading databa
se ... 45%\r(Reading database ... 50%\r(Reading database ... 55%\r(Reading datab
ase ... 60%\r(Reading database ... 65%\r(Reading database ... 70%\r(Reading data
```

Describe the output of this command. Notice how we added the command `state=latest` and placed them in double quotations.

Based from the output of the command, we will finally see that `change=true`. It was able to update `snapt` in the servers. Since `name` and `state` are two different options for the `apt` we have to use quotations so that the second option `state` does not get interpreted as a separate argument/command from the `apt -a`

Task 2: Writing our First Playbook

1. With ad hoc commands, we can simplify the administration of remote servers. For example, we can install updates, packages, and applications, etc. However, the real strength of Ansible comes from its playbooks. When we write a playbook, we can define the state that we want our servers to be in and the place or commands that ansible will carry out to bring to that state. You can use an editor to create a playbook. Before we proceed, make sure that you are in the directory of the repository that we use in the previous activities (`CPE232_yourname`). Issue the command `nano install_apache.yml`. This will create a playbook file called `install_apache.yml`. The `.yaml` is the basic standard extension for playbook files.

When the editor appears, type the following:

```
---
- hosts: all #change if needed
  become: true
  tasks:
    - name: install apache2 package
      apt:
        name: apache2
```

Make sure to save the file. Take note also of the alignments of the texts. Use TAB instead of SPACE BAR for spacing.

2. Run the `yml` file using the command: `ansible-playbook --ask-become-pass install_apache.yml`. Describe the result of this command.

```
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE212... x qimagalan-workstation@workstation: /etc/an... x
qimagalan-workstation@workstation:~/CPE212_Galan$ ansible-playbook install_apache2.yml -b -K -i /etc/ansible/inventory.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [server1]
ok: [Galan_CentOS]
ok: [server2]

TASK [install apache2 package] *****
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
fatal: [Galan_CentOS]: FAILED! => {"changed": false, "cmd": "apt-get update", "msg": "[Errno 2] No such file or directory: b'apt-get'", "rc": 2, "stderr": "", "stderr_lines": [], "stdout": "", "stdout_lines": []}
changed: [server1]
changed: [server2]

PLAY RECAP *****
Galan_CentOS      : ok=1    changed=0    unreachable=0    failed=1
kipped=0    rescued=0    ignored=0
server1          : ok=2    changed=1    unreachable=0    failed=0
kipped=0    rescued=0    ignored=0
server2          : ok=2    changed=1    unreachable=0    failed=0
kipped=0    rescued=0    ignored=0

qimagalan-workstation@workstation:~/CPE212_Galan$
```

3. To verify that apache2 was installed automatically in the remote servers, go to the web browsers on each server and type its IP address. You should see something like this.



4. Try to edit the `install_apache.yml` and change the name of the package to any name that will not be recognized. What is the output?

It shows a status of `ok=1` on the play recap so the servers were reached but it did not change anything because as stated from the detailed error that there was no "ThisCantBeReal" package available, and it is a message from `apt-get update`.

```
Aug 18 10:00
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE212... x qimagalan-workstation@workstation: /etc/an... x

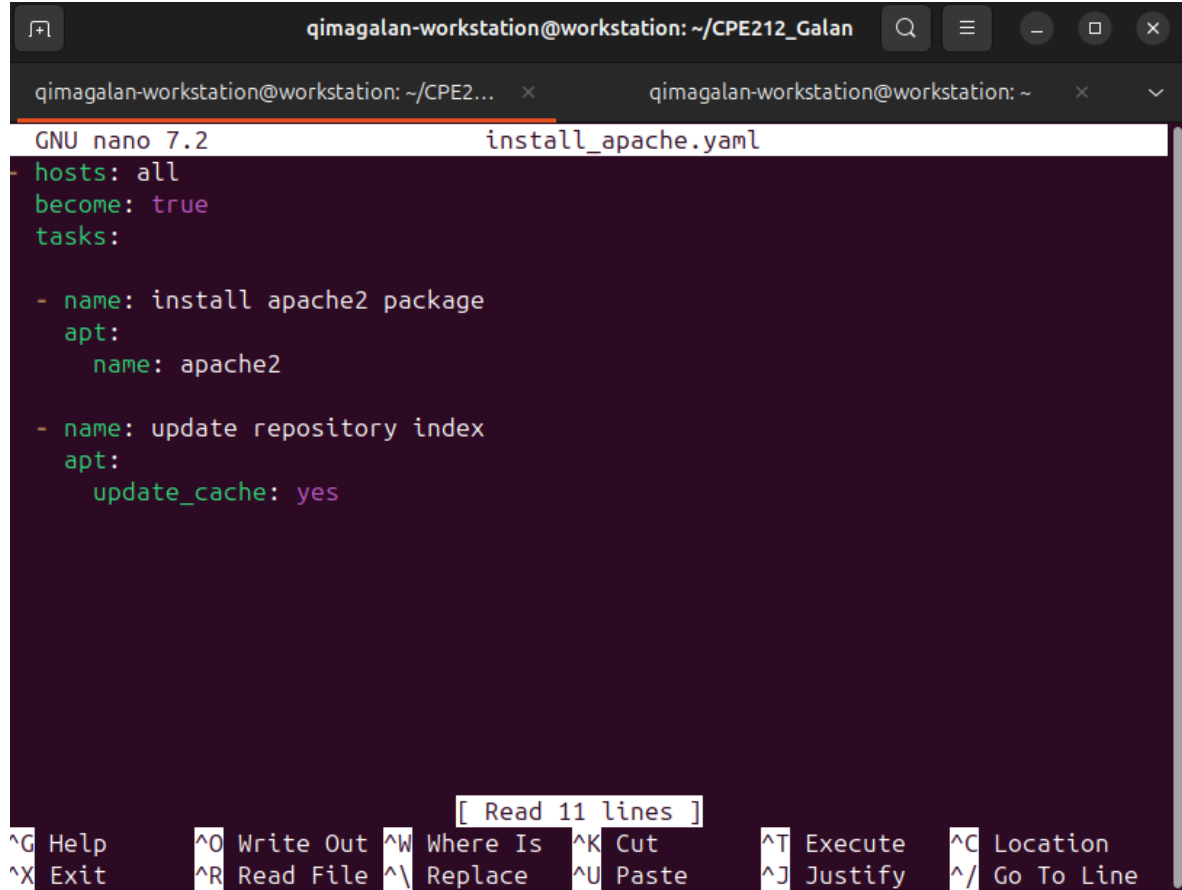
TASK [Gathering Facts] *****
ok: [server1]
ok: [server2]
ok: [Galan_CentOS]

TASK [Test Unknown Package] *****
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
fatal: [Galan_CentOS]: FAILED! => {"changed": false, "cmd": "apt-get update",
sg": "[Errno 2] No such file or directory: b'apt-get'", "rc": 2, "stderr": "",
stderr_lines": [], "stdout": "", "stdout_lines": []}
fatal: [server1]: FAILED! => {"changed": false, "msg": "No package matching 'T
sCantBeReal' is available"}
fatal: [server2]: FAILED! => {"changed": false, "msg": "No package matching 'T
sCantBeReal' is available"}

PLAY RECAP *****
Galan_CentOS      : ok=1    changed=0    unreachable=0    failed=1
kipped=0    rescued=0    ignored=0
server1          : ok=1    changed=0    unreachable=0    failed=1
kipped=0    rescued=0    ignored=0
server2          : ok=1    changed=0    unreachable=0    failed=1
kipped=0    rescued=0    ignored=0

qimagalan-workstation@workstation:~/CPE212_Galan$
```

5. This time, we are going to put additional tasks into our playbook. Edit the *install_apache.yml*. As you can see, we are now adding an additional command, which is the *update_cache*. This command updates existing package-indexes on a supporting distro but not upgrading installed-packages (utilities) that were being installed.



```
qimagalan-workstation@workstation: ~/CPE212_Galan
GNU nano 7.2 install_apache.yml
- hosts: all
  become: true
  tasks:
    - name: install apache2 package
      apt:
        name: apache2
    - name: update repository index
      apt:
        update_cache: yes
```

[Read 11 lines]

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

Save the changes to this file and exit.

6. Run the playbook and describe the output. Did the new command change anything on the remote servers?

Yes it did change the remote servers, we could see that the task name 'update repository index' shows changed: [server2], changed: [server1].

```
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE2... x qimagalan-workstation@workstation: /etc/an... x v
ok: [server2]
ok: [Galan_CentOS]

TASK [install apache2 package] *****
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
fatal: [Galan_CentOS]: FAILED! => {"changed": false, "cmd": "apt-get update", "msg": "[Errno 2] No such file or directory: b'apt-get'", "rc": 2, "stderr": "", "stderr_lines": [], "stdout": "", "stdout_lines": []}
ok: [server2]
ok: [server1]

TASK [update repository index] *****
changed: [server2]
changed: [server1]

PLAY RECAP *****
Galan_CentOS      : ok=1    changed=0    unreachable=0    failed=1    s
kipped=0    rescued=0    ignored=0
server1        : ok=3    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
server2        : ok=3    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0

qimagalan-workstation@workstation: ~/CPE212_Galan$
```

7. Edit again the `install_apache.yml`. This time, we are going to add a PHP support for the apache package we installed earlier.

```
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE212_Galan
GNU nano 7.2 install_apache.yaml
hosts: all
become: true
tasks:
- name: install apache2 package
  apt:
    name: apache2

- name: update repository index
  apt:
    update_cache: yes

- name: add PHP support for apache
  apt:
    name: libapache2-mod-php

[ Read 14 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Save the changes to this file and exit.

8. Run the playbook and describe the output. Did the new command change anything on the remote servers?

Yes it did change the remote servers, we could see that the task name 'add PHP support for apache' shows changed: [server2], changed: [server1].

```
Aug 18 10:09
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE212_Galan x qimagalan-workstation@workstation: /etc/an...
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
fatal: [Galan_CentOS]: FAILED! => {"changed": false, "cmd": "apt-get update", "msg": "[Errno 2] No such file or directory: b'apt-get'", "rc": 2, "stderr": "", "stderr_lines": [], "stdout": "", "stdout_lines": []}
ok: [server2]
ok: [server1]

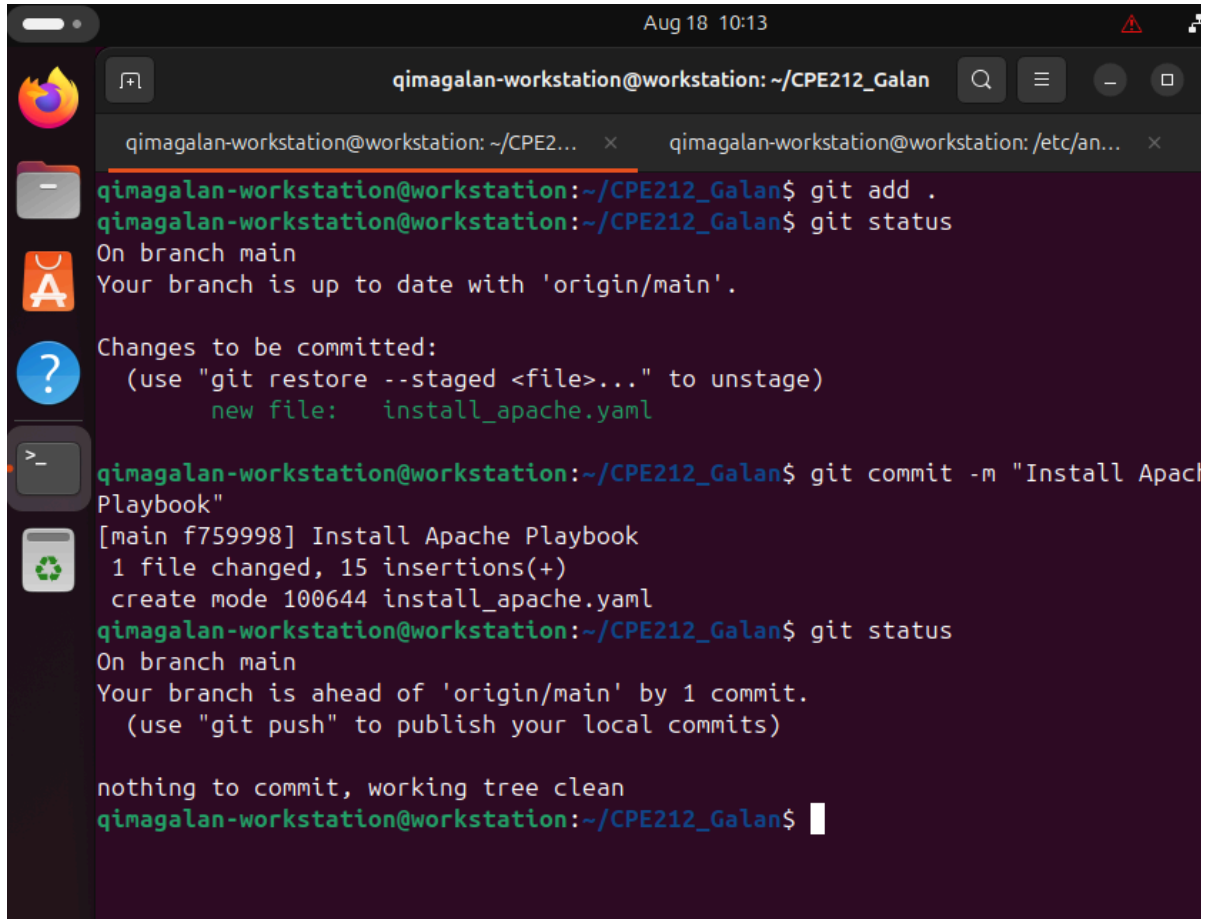
TASK [update repository index] *****
changed: [server2]
changed: [server1]

TASK [add PHP support for apache] *****
changed: [server1]
changed: [server2]

PLAY RECAP *****
Galan_CentOS      : ok=1    changed=0    unreachable=0    failed=1    s
kipped=0    rescued=0    ignored=0
server1         : ok=4    changed=2    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
server2         : ok=4    changed=2    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0

qimagalan-workstation@workstation: ~/CPE212_Galan$
```

9. Finally, make sure that we are in sync with GitHub. Provide the link to your GitHub repository.



```
Aug 18 10:13
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE212_Galan$ git add .
qimagalan-workstation@workstation: ~/CPE212_Galan$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   install_apache.yaml

qimagalan-workstation@workstation:~/CPE212_Galan$ git commit -m "Install Apache Playbook"
[main f759998] Install Apache Playbook
 1 file changed, 15 insertions(+)
 create mode 100644 install_apache.yaml
qimagalan-workstation@workstation:~/CPE212_Galan$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
qimagalan-workstation@workstation:~/CPE212_Galan$
```

```

Aug 18 10:13
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE212_Galan$ git add .
qimagalan-workstation@workstation: ~/CPE212_Galan$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   install_apache.yaml

qimagalan-workstation@workstation:~/CPE212_Galan$ git commit -m "Install Apache Playbook"
[main f759998] Install Apache Playbook
 1 file changed, 15 insertions(+)
 create mode 100644 install_apache.yaml
qimagalan-workstation@workstation:~/CPE212_Galan$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
qimagalan-workstation@workstation:~/CPE212_Galan$

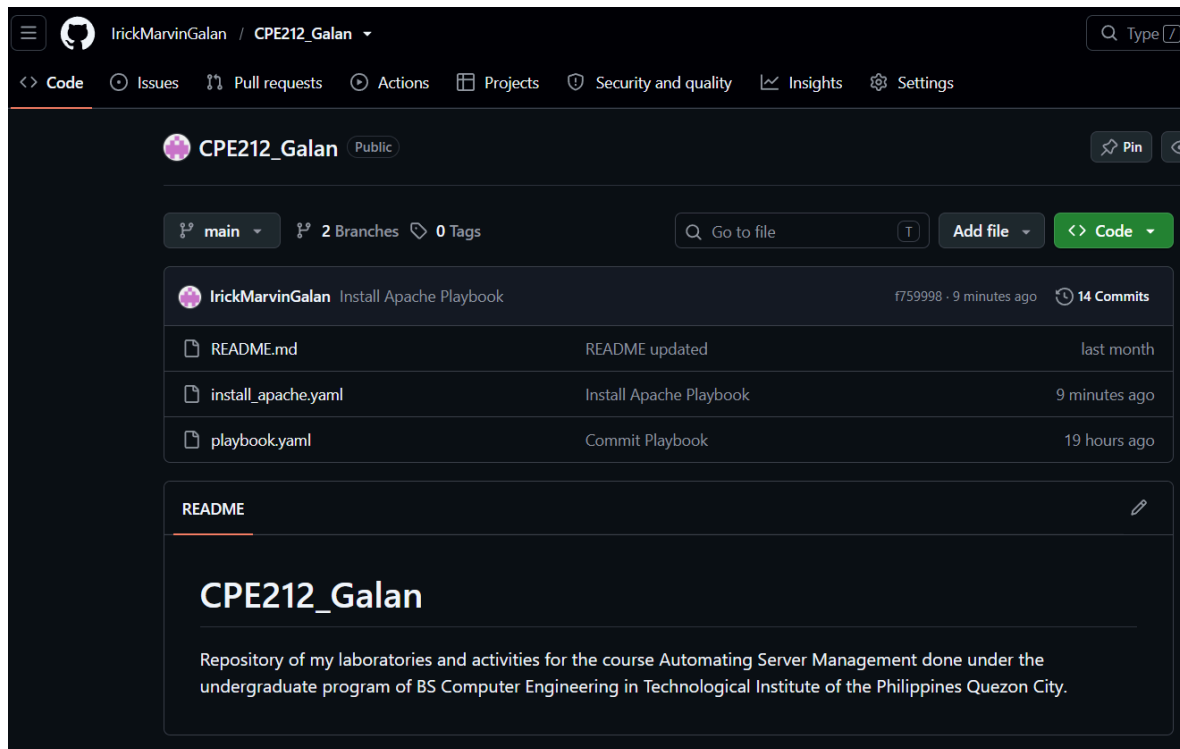
```

Then for the final step issue command, **git push origin main** (Forgot to screenshot)

```

96 clear
97 git push origin main
98 clear
99 history
qimagalan-workstation@workstation:~/CPE212_Galan$

```



Github Link: https://github.com/IrickMarvinGalan/CPE212_Galan

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What is the importance of using a playbook?

A playbook alleviates the burden of having to enter numerous ad hoc commands which will take an effort to do every single time. By using a playbook we could overcome it since it specifies everything from task name, and the metagroup that is targeted by the command. As we advance through this course, we could therefore say that playbooks offer a lot of advantage in automation and scheduled executions since it compresses all commands to execute in a file. If the commands to be executed is routinary, we could simply use a playbook and run it each time.

2. Summarize what we have done on this activity.

In this activity, we primarily looked at privilege escalation using `-become` and `-ask-become-pass`. We used them for ad hoc commands with `apt`, as well as applied them into a playbook. We also take a look at how chained optional arguments for `apt` such as `name` and `state` should be enclosed within quotations to make it function.

Conclusions:

In this activity, I have learned that using `-b -K` for simplicity of privilege escalation is an important aspect of executing ansible playbooks especially for commands that change the system like `apt`. I also observed in this activity the parallel between using the ad hoc `-m <command>` format and how it actually translates to the playbook, because I noticed that how the first argument after `-m` usually becomes the command directly below the `'name:'` block in

playbooks, and the name= translates into a 'name:' block under the 'apt:' block which gives us a clearer idea how the command hierarchy translates from ad hoc to ansible playbooks. This particular observation made using ansible playbooks more intuitive. Lastly, the changed status on the PLAY RECAP is also important since it shows us if the executed playbook did modify the servers. Being able to read the PLAY RECAP is valuable since it summarizes whether the host is reachable, if the command is ignored, and many other statuses which we did not get to see yet in this activity.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

No AI was used for this activity.

Purpose of Use:

No AI was used for this activity.

Extent of Use:

No AI was used for this activity.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.